MASTER OF COMPUTER APPLICATIONS

(MCA)

(MC - 401)
PROJECT GUIDELINES
(July - 2026)

GANGA GLOBAL INSTITUTE OF MANAGEMENT STUDIES
BEGUSARAI, BIHAR

CONTENTS

MESSAGE FROM THE PROJECT CO-ORDINATOR

The Master of Computer Applications (MCA) programme prepares the students to take up positions as Systems Analysts, Systems Designers, Software Engineers, Programmers and Project Managers in any field related to information technology. We had therefore imparted you the comprehensive knowledge covering the skills and core areas of computer science courses with equal emphasis on the theory and practical. The MCA students are expected to spend last semester (4^{th} semester) working on a project preferably in a software industry or any research organization.

The theoretical background of various courses provides you the necessary foundation, principles, and practices to develop effective ways to solve computing problems. The hands on experience gained from the practical courses provides you the knowledge to work with various operating systems, programming languages, software tools and testing tools.

The objective of the MCA project work is to develop quality software solution. During the development of the project, you should involve in all the stages of the software development life cycle like requirements engineering, systems analysis, systems design, software development, testing strategies and documentation with an overall emphasis on the development of reliable software systems. The primary emphasis of the project work is to understand and gain the knowledge of the principles of software engineering practices, so as to participate and manage a large software engineering projects in future.

Approval of the project proposal is mandatory to continue and submit the project work. Prepare your project proposal strictly as per guidelines. Disapproval of project proposal leads to loss of your valuable time. To avoid this loss take your proposal preparation very seriously and consult for every point on which you have doubt, with your project guide/supervisor.

You are advised to take this project work **very seriously**, as these efforts will be considered as 6- months experience in most of the software companies. Topics selected should be complex and large enough to justify as a MCA project. The project should be genuine and original in nature and should not be copied from anywhere else. If found copied, the project report will be forwarded to the Exam Discipline Committee of the University as an Unfair means case for necessary action. In case of project resubmission, please confirm the fees and other details with the College. Students should strictly follow and adhere to the project guidelines.

I wish you all the success.

MCA Project Coordinator

I CALENDAR FOR THE MCA PROJECT

Sl.No.	*Topic*	*Date*
1.	Submission of a guide's bio-data and project proposal at the following address: **Project Coordinator**	**1st May to 7th May**
2.	Approval of Project	**20 days** after the project proposal is received.
3.	Submission of the Project Report (**Three copy (Student/University/Project Co-ordinator)**) **in bound form to:** **Project Co-Ordinator/ University**	**15st May to 4th June**
4.	Viva-Voce to be conducted	**As Per University Calendar**

II. PROFORMA FOR THE APPROVAL OF MCA PROJECT PROPOSAL

(Note: All entries of the proforma of approval should be filled up with appropriate and complete information. Incomplete proforma of approval in any respect will be summarily rejected.)

Project Proposal No :…………………..
(for office use only)

Registration No.: ……………………
E-mail: ……….……….……….…...
Mobile/Tel No.: …..…………………

1. Name and Address of the Student: ………………………..…………………………………….

   ……………………………………………………………….

2. Title of the Project***: .………..…………………………………………………….

3. Name and Address of the Guide: …..………………………………………………………….

   ……………………………………………………………….

4. Educational Qualification of the Guide: (Attach bio-data also)

   Ph.D* ☐ M.Tech.* ☐ B.E*/B.Tech.* ☐ MCA ☐ M.Sc.* ☐

   (*in Computer Science / IT only)

5. Working / Teaching experience of the Guide**:……………………………………………………………

   …….

(***Note: At any given point of time, a guide should not provide guidance for more than 5 MCA students of GGIMS*)

6. Software used in the Project***:………………………………………………………..
   (*** Please refer to section VIII of these guidelines)

7. Is this your first submission? ☐ Yes ☐ No

Signature of the Student
Date: …………………

Signature of the Guide
Date: …………………….

For Office Use Only

☐ Approved ☐ Not Approved

Name:………………… ………..............

…………………………………………….
Signature, Designation, Stamp of the Project Proposal Evaluator
Date:…………………….

<u>Suggestions for reformulating the Project:</u>

Ensure that you include the following while submitting the Project Proposal:

1. **Proforma for Approval of Project Proposal duly filled and signed by both the student and the Project Guide with date.**

2. **Bio-data of the project guide with her/his signature and date. Please also attach the certified copy of proof of identity of guide.**

3. **Synopsis of the project proposal (14-18 pages).**

Note:

i. ***At any given point of time, a guide should not provide guidance for more than 5 MCA students of GGIMS.***

ii. ***If your project proposal is Not Approved then suggestions given for reformulating the project must be incorporated in the new project proposal.***

iii. ***If your project proposal is approved then suggestions given for reformulating the project must be incorporated in the final project otherwise the project may be rejected at any stage of evaluation.***

iv. ***On approval of your project proposal, you must perform a thorough analysis, design, implementation and testing of your project.***

v. ***Violation of the project guidelines will lead to the rejection of the project at any stage.***

A photocopy of the complete Project Proposal (along with Project Performa, Project Synopsis, Bio data of the guide) submitted to your College, should be retained by the student for future reference.

III INTRODUCTION AND OBJECTIVES

The Project work constitutes a major component in most professional programmes. It needs to be carried out with due care, and should be executed with seriousness by the students. The project work is not only a partial fulfilment of the MCA requirements, but also provides a mechanism to demonstrate your skills, abilities and specialisation. The project work should compulsorily include the software development. Physical installations or configuring the LAN/WAN or theoretical projects or study of the systems, which doesn't involve s/w development, are strictly not allowed.

OBJECTIVES

The objectives of the project is to help the student develop the ability to apply theoretical and practical tools/techniques to solve real life problems related to industry, academic institutions and research laboratories. After the completion of this project work, the student should be able to:

- Describe the Systems Development Life Cycle (SDLC).
- Evaluate systems requirements.
- Complete a problem definition.
- Evaluate a problem definition.
- Determine how to collect information to determine requirements.
- Perform and evaluate feasibility studies like cost-benefit analysis, technical feasibility, time feasibility and Operational feasibility for the project.
- Work on data collection methods for fact finding.
- Construct and evaluate data flow diagrams.
- Construct and evaluate data dictionaries.
- Evaluate methods of process description to include structured English, decision tables and decision trees.
- Evaluate alternative tools for the analysis process.
- Create and evaluate such alternative graphical tools as systems flow charts and state transition diagrams.
- Decide the S/W requirement specifications and H/W requirement specifications.
- Plan the systems design phase of the SDLC.
- Distinguish between logical and physical design requirements.
- Design and evaluate system outputs.
- Design and evaluate systems inputs.
- Design and evaluate validity checks for input data.
- Design and evaluate user interfaces for input.
- Design and evaluate file structures to include the use of indexes.
- Estimate storage requirements.
- Explain the various file update processes based on the standard file organizations.
- Decide various data structures.
- Construct and evaluate entity-relationship (ER) diagrams for RDBMS related projects.
- Perform normalization for the un-normalized tables for RDBMS related projects.

- Decide the various processing systems to include distributed, client/server, online and others.
- Perform project cost estimates using various techniques.
- Schedule projects using both GANTT and PERT charts.
- Perform coding for the project.
- Documentation requirements and prepare and evaluate systems documentation.
- Perform various systems testing techniques/strategies to include the phases of testing.
- Systems implementation and its key problems.
- Generate various reports.
- Be able to prepare and evaluate a final report.
- Brief the maintenance procedures and the role of configuration management in operations.
- To decide the future scope and further enhancement of the system.
- Plan for several appendices to be placed in support with the project report documentation.
- Work effectively as an individual or as a team member to produce correct, efficient, well-organized and documented programs in a reasonable time.
- Recognize problems that are amenable to computer solutions, and knowledge of the tools necessary for solving such problems.
- Develop of the ability to assess the implications of work performed.
- Get good exposure and command in one or more application areas and on the software
- Develop quality software using the software engineering principles.
- Develop of the ability to communicate effectively.

In case, you are using Object Oriented Approach for your project development, draw different UML diagrams to represent system analysis and design.

\

IV TYPE OF THE PROJECT AND ELIGIBILITY CRITERIA OF THE PROJECT GUIDE

Type of the Project

The majority of the students are expected to work on a real-life project preferably in some industry/ Research and Development Laboratories/Educational Institution/Software Company. Students are encouraged to work in the areas listed at the end. However, it is ***not mandatory*** for a student to work on a real-life project. The student can formulate a project problem with the help of her/his Guide and submit the project proposal of the same. **Approval of the project proposal is mandatory**. If approved, the student can commence working on it, and complete it. Use the latest versions of the software packages for the development of the project.

Eligibility criteria of a Project Guide

1. A person having Ph.D./ M.Tech. in Computer Science with a minimum of one year of experience.

Or

2. A person having B.E./B.Tech. (Computer Science), MCA, M.Sc. (Computer Science) with minimum 2 years' experience, preferably in software development.

Steps involved in the project work

The complete project work should be done by the student only. The role of guide should be about guidance wherever any problem encounters during project. The following are the major steps involved in the project, which may help you to determine the milestones and regulate the scheduling of the project:

- Select a topic and a suitable guide.
- Prepare the project proposal in consultation with the project guide.
- Submit the project proposal along with the necessary documents.
- Receipt of the project approval from the College.
- Carry out the project-work.
- Prepare the project report.
- Submit the project report to the Project Coordinator.
- Appear for the viva-voce as per the intimation by the University.

Communication of the approval

Communication regarding the project Approval/Non-approval will be communicate to you within four to six weeks after the receipt of the project proposal by the college concerned. In case you do not receive any communication from your college within four to six week after last date of submission of project proposal/synopsis, you are advised to contact your college.

Resubmission of the project proposal in case of non-approval

In case of non-approval, the suggestions for reformulating the project will be communicated to you. The revised project synopsis along with a new proforma, should be re-submitted along with a copy of the earlier synopsis and non-approval project proposal proforma in the **next slot.** If the student wants to change the project topic or software or the project guide, s/he may do so and can submit a fresh project proposal. **The revised project proposal should be sent along with the original copy/ photocopy of the non-approved proforma of the earlier submitted proposal, to the College.**

Resubmission of MCA project in case of failed students

If the student is unsuccessful in the project, s/he should 're-do' the whole cycle, right from the submission of the project proposal. Students are advised to select a **new topic** for the project and should prepare and submit the project proposal to the College concerned as per the project guidelines. There are no separate slots for the submission of the project synopsis/project reports for the failed students. Respective submissions of the project synopsis and the project reports should be done strictly as per the "Calendar for the MCA project" given in the project guidelines.

V POINTS TO REMEMBER WHILE PREPARING THE PROJECT PROPOSAL

1. Project Proposal Formulation

- **The project proposal should be prepared in consultation with your guide**. The project proposal should clearly state the project objectives and the environment of the proposed project to be undertaken. **The project work should compulsorily include the software development.** The project proposal should contain complete details in the following form:

- Proforma for Approval of Project Proposal duly filled and signed by both the student and the Project Guide with date.
- Bio-data of the project guide with her/his signature and date.
- Synopsis of the project proposal (15-20 pages) covering the following aspects:

  (i) Title of the Project.
  (ii) Introduction and Objectives of the Project.
  (iii) Project Category (RDBMS/OOPS/Networking/Multimedia/Artificial Intelligence/Expert Systems etc.).
  (iv) Tools/Platform, Hardware and Software Requirement specifications.
  (v) Problem Definition, Requirement Specifications (Detailed functional Requirements and Technical Specifications), Project Planning and Scheduling (Gantt chart and PERT chart).
  (vi) Scope of the solution.
  (vii) Analysis (Data Models like 0, 1 and 2 level DFDs, Complete ER Diagrams with cardinality, Activity Diagram, Class Diagrams, State Diagrams etc. as per the project requirements).
  (viii) A complete Database and tables detail with Primary and Foreign keys, and proper constraints in the fields (as per project requirements)
  (ix) A complete structure which includes:
    - Number of modules and their description to provide an estimation of the student's effort on the project. Along with process logic of each Module.
    - Data Structures as per the project requirements for all the modules.
    - Process Logic of each module.
    - Implementation methodology
    - List of reports that are likely to be generated.

(x) Overall network architecture (if required for your project)
(xi) Implementation of security mechanisms at various levels
(xii) Future scope and further enhancement of the project.
(xiii) Bibliography

2. All entries of the proforma of approval should be filled up with appropriate and complete information. Incomplete approval-proforma in any respect will be summarily rejected.

3. A **photocopy of the complete Project Proposal** (along with Project Proforma, Project Synopsis, Bio-data of the guide) submitted to your college, should be retained by the student for future reference.

4. The evaluated project proposal proforma along with the details of Approved/Disapproved will be sent to the student within 4-6 weeks after the proposal is received at college concerned. In case if it is disapproved, the suggestions for reformulating the project will be communicated to the student. Revised project proposal proforma, synopsis, bio-data of the guide with her/his signature on it, should be sent along with the original copy/photocopy of the non-approved proforma of the earlier project proposal, to the college Concerned.

5. The project is a part of your final semester (4^{th} semesters) curriculum. Students are eligible to submit the project proposals as per the calendar.

6. Violation of the project guidelines will lead to the rejection of the project at any stage.

VI POINTS TO REMEMBER WHILE PREPARING THE PROJECT REPORT

1. Project Report Formulation:

The project report **should** contain the following:

(i) Original copy of the Approved Proforma and Project Proposal.
(ii) Bio-data of the guide with her/his signature and date.
(iii) Certificate of Originality.
(iv) Project documentation.
(v) A **Pen Drive** consisting of the executable file(s) of the complete project should be attached on the last page of the project report. In no case, it should be sent separately.

2. The **project documentation** may be about 100 to 125 pages (excluding coding). The project documentation details should not be too generic in nature. Appropriate project report documentation should be done, like, **how you have done the analysis, design, coding, use of testing techniques/strategies, etc.,** ***in respect of your project***. To be more specific, whatever the theory in respect of these topics is available in the reference books should be avoided as far as possible. **The project documentation should be in respect of your project only.** The project documentation should include the topics given below. Each and every component shown below carries certain weightage in the project report evaluation.

- Table of Contents/Index with page numbering
- Introduction/Objectives
- System Analysis
  - Identification of Need
  - Preliminary Investigation
  - Feasibility Study
  - Project Planning
  - Project Scheduling (PERT Chart and Gantt Chart both)
  - Software requirement specifications (SRS)
  - Software Engineering Paradigm applied
  - Data models (like DFD), Control Flow diagrams, State Diagrams/Sequence diagrams, Entity Relationship Model, Class Diagrams/CRC Models/Collaboration Diagrams/Use-case Diagrams/Activity Diagrams depending upon your project requirements
- System Design
  - Modularisation details
  - Data integrity and constraints
  - Database design, Procedural Design/Object Oriented Design
  - User Interface Design
  - Test Cases (Unit Test Cases and System Test Cases)
- Coding
  - SQL commands for (i) database creation (along with constraints), (ii) data insertion in tables and (iii) access rights for different users.
  - Complete Project Coding
  - Comments and Description of Coding segments

  - ♦ Standardization of the coding
  - ♦ Code Efficiency
  - ♦ Error handling
  - ♦ Parameters calling/passing
  - ♦ Validation checks
- ♦ Testing
  - ♦ Testing techniques and Testing strategies used
  - ♦ Testing Plan used
  - ♦ Test reports for Unit Test Cases and System Test Cases
  - ♦ Debugging and Code improvement
- ♦ System Security measures (Implementation of security for the project developed)
  - ♦ Database/data security
  - ♦ Creation of User profiles and access rights
- ♦ Cost Estimation of the Project along with Cost Estimation Model
- ♦ Reports (sample layouts should be placed)
- ♦ Future scope and further enhancement of the Project
- ♦ Bibliography
- ♦ Appendices (if any)
- ♦ Glossary.
- ♦ **Should attach a Pen Drive containing the executable file(s) of the complete project.**

3. The project report should normally be printed with single line spacing on A4 paper (one side only). All the pages, tables and figures must be numbered. Tables and figures should contain titles.

4. If any project report is received in the absence of the approved project proposal proforma (in original), project synopsis, bio-data of the guide with her/his signature on it, certificate of originality and Pen Drive it will be summarily rejected and returned to the student for compliance.

5. Throughout the project report, the title of the project should be the same as per the approved synopsis. Signature of the Project Guide in the Certificate of Originality should match with the signature in the Project Proposal proforma also.

6. **Only one copy of** the **original project report** in the bound form along with the Pen Drive (containing the executable file(s) of the project should be enclosed in the last page) is to be submitted to the college by date mentioned in the Calendar for the project. One photocopy of the same Project Report and the Pen Drive containing the executable file(s) must be retained by the student, which should be produced before the examiner at the time of viva-voce.

7. A photocopy of the project report is **not acceptable** for submission.

8. **Preferably**, **not more than one student is permitted to work on a project**. However, in case a project is comprehensive enough that requires one human-year or more time for its completion, then as per requirements of six human-months per student, at most two students

may work on the same project. **In this regard, prior recommendation is mandatory and must be obtained from the MCA Project Coordinator** by sending the complete synopsis by both the students along with a hand-written application.

9. If 2 students have been allowed to work on a project, the project synopsis and project reports by them must include only different modules undertaken/worked upon **individually.** Each student must submit a **separate** project proposal and a separate project reports related to her/his modules. **Completely identical project synopsis and/or project reports are not allowed**. Only introductory and possibly concluding remarks may be similar or common. Each student has to undergo all the phases/stages of the software project development life cycle. In this case, both the students must attach **the prior recommendation obtained from the MCA Project Coordinator. A single copy of the project synopsis and/or project report comprising of work of two or more students shall not be entertained. Violation of these project guidelines may lead to the rejection of the project at any stage.**

10. Student should be involved in each and every phase of Project Development. If it is found that student is not involved in any phase for example coding phase, it may lead to the rejection/disqualifying of the project at any stage.

11. Title of the project should be kept the same throughout the project.

VII ASSESSMENT GUIDELINES FOR PROJECT EVALUATION

Each and every component of the project work and the viva voce carries its own weightage, so the student needs to concentrate on all the sections given in the project report formulation.

In this section, we have given a few general parameters as checkpoints for the assessment of any software development project. You can note these points and emphasise them during the project report preparation and examination. Basically, assessment will be based on the quality of your report, the technical merit of the project and the project execution. Technical merit attempts to assess the quality and depth of the intellectual effort, you have put into the project. Project execution is concerned with assessing how much work you have put in.

Analysis

In Project planning include cost estimation and project scheduling. The Cost and efforts estimation is to be done with the help of functional point analysis or other similar methods. The project scheduling is identified with:

(i) PERT chart: Proper decomposition of stages, and
(ii) Gantt chart: Time, line structure and validity of chart.

You may know that the software requirement specification (SRS) document is one of the important documents of your project. The indicators for SRS document is whether you have used some standardization like IEEE standards or any other international standard, or whether your SRS has a proper structure based on sound software engineering concepts as given in unit 3 or it is given in a running text. Project analysis for DBMS/Application development projects should contain the ER diagram, Data Flow Diagram and Data Dictionary, so you should include these with the following requirements. However, for other categories of project you should prepare class diagrams, behaviour model and/or state transition diagram and details of various data structures used.

- The Entity Relationship diagram (ER Diagram) should have:
    - Proper symbol of attributes, entities, relationships, cardinality mentioned, and
    - Relationship of ER diagram to SRS with strong association
    - Data Flow Diagram (DFD) should have:

    - All Data flow should be levelled and should have proper input and output.
    - Relationship of data flow to data dictionary Context Diagram, Level 1 and Level 2.

- **Data Dictionary:** It should explain each entity and relationship in ER diagram and data flow in DFD.

Design

Project design should include the desired features and operations in detail, including user interface design, program structure, schema design and normalised tables and data integrity and constraints. You should include them with the requirements given below:

- **Program Structure:** It should have the proper modularisations of software and specification of each module.

- **Schema Design and Normalised Tables:** Normalise the table to minimum 3NF. If any demand of Demoralisations clearly explain the reasons.
- **Data Integrity and Constraints:** Explain the referential diagram. Define entity integrity, which should include keys, alternate keys and other keys, value constraints and ranges.
- **Procedural Design:** Explain using Flowchart / Pseudo code / PDL.
- **User Interface Design:** Coherence with dataflow and processor; Consistency of interface and naming convention. Validation checks should be kept wherever necessary.
- **Architecture:** Program architecture and explanation on suitability of data structure used.

Coding

Coding phase of software development includes different activities like refining the algorithms for individual components, transferring the algorithms into a programming language (coding), translating the logical data model into a physical one and compiling and checking the syntactical correctness of the algorithm with these activities. You should include the comments and description in code, use the standardisation in coding, use the methodology for error handling and security implementation. These parameters ensure software quality and reliability. You should include them with the requirements given below:

- **Comments and Description:** Should have comments with functional description which include the input, output, total function calls to/from other functions, function parameters, description of main variables, Data type, logic description, etc.
- **Standardisation of Coding:** Use of naming convention of variable and functions, nested depth, naming constant, use of data structure and style.
- **Error Handling:** Explain exceptions handling and conditional checking.
- **Parameter passing and calling:** Check the technique used for this purpose, how it optimises the coding.
- **Security Mechanisms:** Maintain confidentiality, integrity and authorisation according to the requirement and needs of the system. Also maintain database level security, use of Views, use of revoke and grant, user and access rights and ensure steps taken against hacking of the system.

Testing

Testing is a process of devising a set of inputs to a given piece of software that will cause the software to exercise some portion of its code. The developer of the software can then check if the results produced by the software are in accordance with his or her expectations. It includes, number of activities such as correcting syntactically and semantically erroneous system components,

detecting as many errors as possible in the software system, and assuring that the system implementation fulfils system specification.

It ensures the quality, efficiency and reliability of the software, which is measured by the testing methodology and techniques used for unit, integrated, system testing etc.

The testing should not be too generic containing only definitions. You should give all the test case designs, reports and results of test cases for unit, integrated, system testing etc. How debugged your code is and what actions you have taken too improve the code, must, be explained. Good testing can be measured by criteria such as correctness, reliability, user friendliness, maintainability, efficiency and portability of software.

System Security Measures

The student should clearly emphasize the various levels of security measures implemented in the project.

Report Generation

The project report should include the various sample reports for ready reference.

Cost Estimation of the project

The student needs to incorporate the estimated cost of the project using the suitable mechanism/model given in the Software Engineering.

Screen Layouts/Screen Shots/Screen dumps

Screen dumps for various screens/user interfaces should also be placed in a proper order in the project report for ready reference.

Organisation of the Project Report

Report organisation improves the professional attitude of writing reports. You should emphasise on the proper binding of the project report, the cover page, page numbering, organisation of content, and proper printout of text and images.

Viva Voce

In this component of evaluation students get chance to present their knowledge and skill to the expert. Other than the questions related to the project related areas and the courses concerned, student may be requested to show the demo of the project. Also, you may be told to write the portions of the code for a problem to demonstrate her/his coding capabilities.

While appearing for the viva-voce, along with the project report the student should needs to carry the identical copy of the Pen Drive of the executable file(s) which s/he submitted at the time of project report.

Project Evaluation

The Project Report is evaluated for 500 marks and the viva-voce. Viva-voce is compulsory and forms part of evaluation. A student in order to be declared successful in the project must secure **40% marks in each component (i) Project Evaluation and (ii) Viva- voce**. Pass in both the components is compulsory. If a student submitted the project report as per the schedule and fails to attend viva, her/his Project will remains incomplete and should contact the College concerned.

VIII SOFTWARE AND BROAD AREAS OF APPLICATION

FRONT END / GUI Tools	Visual Basic, Power Builder, X-Windows (X/lib, X/motif, X/Intrinsic), Oracle Developer 2000,VC++, Jbuilder , NetBeans
RDBMS/BACK END	Oracle, Ingres, Sybase, Progress, SQL Plus, Versant, MY SQL, SQL Server, DB2, Point base
LANGUAGES	C, C++, Java, VC++, C#, eclipse
SCRIPTING LANGUAGES	PERL, SHELL Scripts (Unix), TcL/TK,
RDBMS/BACK END	Oracle, Ingres, Sybase, Progress, SQL Plus, Versant, MY SQL, SQL Server, DB2
.NET Platform	VB.Net, C#. Net, Visual C#. Net, ASP.Net
MIDDLE WARE (COMPONENT) TECHNOLOGIES	COM/DCOM, Active-X, EJB, WINCE, MSMQ, BEA, MessageQ, MTS, CICS
UNIX INTERNALS	Device Drivers, RPC, Threads, Socket programming
ARCHITECTURAL CONCEPTS	CORBA, TUXEDO, MQ SERIES
INTERNET TECHNOLOGIES	DHTML, Java script, VB Script, Perl & CGI script, Java, Active X, RMI, CORBA, SWING, JSP, ASP, XML, EJB, Java Beans, Servlets, Visual Age for JAVA, UML, VRML, WML, Vignette, EDA, Broadvision, Ariba, iPlanet, ATG, BigTalk, CSS, XSL, Oracle ASP server, AWT, J2EE, LDAP, ColdFusion, Haskell 98, PHP,NetBeans
NETWORK/WIRELESS TECHNOLOGIES	Blue Tooth, 3G, ISDN, EDGE
REALTIME OPERATING SYSTEM/ EMBEDDED SKILLS	QNX, LINUX, OSEK, DSP, VRTX, RTXC, Nucleus
OPERATING SYSTEMS	WINDOWS 2000/ME, WINDOWS NT, WINDOWS XP, UNIX, LINUX, IRIX, SUN SOLARIS, HP/UX, PSOS, VxWorks, AS400, AIX, DOS
APPLICATION AREAS	Financial / Insurance / Manufacturing / Multimedia / Computer Graphics / Instructional Design/ Database Management System/ Internet / Intranet / Computer Networking-Communication Software development/ E-Commerce/ ERP/ MRP/ TCP-IP programming / Routing protocols programming/ Socket programming.

GANGA GLOBAL INSTITUTE OF MANAGEMENT STUDIES

BEGUSARAI, BIHAR

Project Trainee Letter (MC-401)

Date:

This is to certify that Mr / Ms____ with Registration No…………………………………………. is a fourth semester student of the Master of Computer Applications (MCA), (GGIMS), and is required to do a six months MCA project work in his/her final year. Her/His project must be undertaken in a software development Organization/ Industry/Research Laboratory under the supervision of a guide, preferably from the same organization with the educational qualifications and experience mentioned in the MC-401 project guidelines. During her/his course of study, the student has studied and gained knowledge on various Computer Science courses such as Problem Solving and Programming, Systems Analysis and Design, Internet Concepts and Web Design, Data Structures, Design and Analysis of Algorithms, Computer Organization, Database Management Systems, Operating Systems, Object Oriented Analysis and Design, Discrete Mathematics, Accountancy and Financial Management, Computer Networks, Software Engineering, Numerical and Statistical Computing, Parallel Computing, Artificial Intelligence and Knowledge Management, Principles of Management and Information Systems, Computer Graphics and Multimedia. S/he has hands on experience in C programming, Assembly Language Programming, Internet Technologies, Oracle / My SQL, JAVA, TALLY, UNIX O/S, Linux O/S etc. Also the student has executed a Mini-Project in the fourth semester using S/W Engineering principles and also studied a course on Communication Skills in the first semester. S/he may please be given a chance to work in your esteemed organisation and complete her/his project work. I ensure you a sincere and quality output from him. The experience gained by this project work, not only benefit the student to partially fulfil the requirements of the MCA of GGIMS, but also lay a foundation for her/his future career.

Looking forward to your positive response, support and cooperation.

Signature, Project Coordinator

X. CERTIFICATE OF ORIGINALITY

This is to certify that the project report entitled ____

submitted to **GANGA GLOBAL INSTITUTE OF MANAGEMENT STUDIES** in partial fulfilment of the requirement for the award of the degree of **MASTER OF COMPUTER APPLICATIONS (MCA) ,** is an authentic and original work carried out by Mr. / Ms.

……………………………………………………………………………………………………

With Registration no.______________________under my guidance.

The matter embodied in this project is genuine work done by the student and has not been submitted whether to this University or to any other University / Institute for the fulfilment of the requirements of any course of study.

……………………….	...…………………….
Signature of the Student:	Signature of the Guide
Date: ………………..	Date: …………………
Name and Address of the student	Name, Designation and Address of the Guide:
………………………..	………………………..
………………………..	………………………..
………………………..	………………………..
………………………..	………………………..

Registration No…………

XI SAMPLE COVER PAGE OF THE PROJECT REPORT

GANGA GLOBAL INSTITUTE OF MANAGEMENT STUDIES

MC-401
TITLE OF THE PROJECT

by

Student's Name:
Registration No:

Under Guidance
of
Project Guide's Full Name

Submitted to the Computer and Information Sciences Department,

GGIMS in partial fulfilment of the requirements

for the award of the degree

Master of Computer Applications (MCA)

Year of Submission